



Schwarz method in time for internal controlled wave problem

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1. Minimization problem

Minimize the functional

$$J(u) = \frac{1}{2} \int_{\Omega \times [0, T]} |y(x, t) - \hat{y}(x, t)|^2 dx dt + \frac{\gamma}{2} \int_{\Omega} |y(x, T) - \hat{z}(x)|^2 dx + \frac{\alpha}{2} \int_{\Omega \times [0, T]} |u(x, t)|^2 dx dt, \quad (1)$$

where u is a control and y is the solution of system

$$\begin{cases} y_{tt} - \Delta y = u, & (x, t) \in \Omega \times [0, T], \\ y(x, t) = 0, & x \in \partial\Omega, t \in [0, T], \\ y(x, 0) = y^0(x), y_t(x, 0) = y^1(x), & x \in \Omega. \end{cases} \quad (2)$$

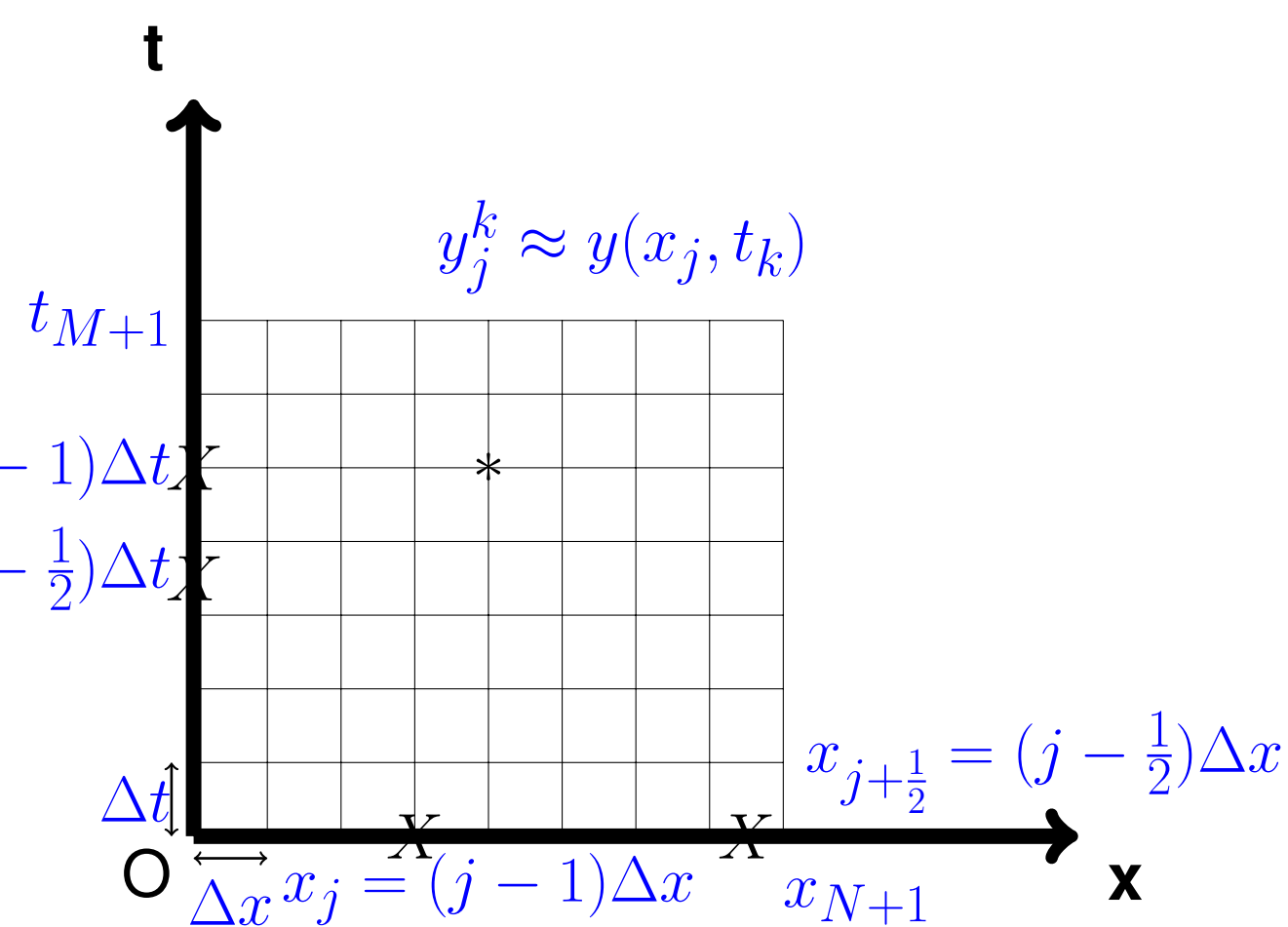
Theorem 1.1. The minimization problem (1) with the constraint (2) is well-posed.

Optimality system (Coupled forward and backward wave equation):

$$\begin{cases} y_{tt} - \Delta y = \frac{\lambda}{\alpha}, \\ \lambda_{tt} - \Delta \lambda + y - \hat{y} = 0, \text{ (adjoint equation)} \\ \gamma(y(\cdot, T) - \hat{z}) - \lambda_t(\cdot, T) = 0, \\ \lambda(\cdot, T) = 0, \\ \lambda = \alpha u, \\ \lambda = 0, \end{cases} \quad \text{on } \Gamma = \partial\Omega, \quad (3)$$

2. Discrete optimality system

Uniform mesh with $\Delta t = \frac{T}{M}$ and $\Delta x = \frac{1}{N}$, $k \in [1, M+1]$ and $j \in [1, N+1]$.



Using the finite volume approach (explicit in time), discretization scheme corresponding with the optimality system (3).

$$\begin{cases} \frac{y_j^{k+1} - 2y_j^k + y_j^{k-1}}{(\Delta t)^2} - \frac{y_{j+1}^k - 2y_j^k + y_{j-1}^k}{(\Delta x)^2} - \frac{1}{\alpha} \lambda_j^k = 0, \\ \frac{y_j^2 - y_j^1}{\Delta t} - \frac{(\Delta t) y_{j+1}^1 - 2y_j^1 + y_{j-1}^1}{(\Delta x)^2} - \frac{\Delta t}{2\alpha} \lambda_j^1 \\ = \frac{1}{\Delta x} \int_{x_j - \frac{\Delta x}{2}}^{x_j + \frac{\Delta x}{2}} y_t(x, 0) dx, \\ \frac{\lambda_j^{k+1} - 2\lambda_j^k + \lambda_j^{k-1}}{(\Delta t)^2} - \frac{\lambda_{j+1}^k - 2\lambda_j^k + \lambda_{j-1}^k}{(\Delta x)^2} + y_j^k = \hat{y}_j^k, \\ \frac{\lambda_j^{M+1} - \lambda_j^M}{\Delta t} + \frac{(\Delta t) \lambda_{j+1}^{M+1} - 2\lambda_j^{M+1} + \lambda_{j-1}^{M+1}}{(\Delta x)^2} \\ = \frac{(\Delta t)}{2} (y_j^{M+1} - \frac{1}{\Delta x} \int_{x_j - \frac{\Delta x}{2}}^{x_j + \frac{\Delta x}{2}} \hat{y}(x, t_{M+1}) dx) \\ + \gamma (y_j^{M+1} - \frac{1}{\Delta x} \int_{x_j - \frac{\Delta x}{2}}^{x_j + \frac{\Delta x}{2}} \hat{z}(x) dx), \end{cases} \quad (4)$$

$$\text{Set } Y^k = \begin{pmatrix} y_1^k \\ y_2^k \\ \vdots \\ y_{N+1}^k \end{pmatrix} \text{ and } \Lambda^k = \begin{pmatrix} \lambda_1^k \\ \lambda_2^k \\ \vdots \\ \lambda_{N+1}^k \end{pmatrix}.$$

$$\text{Arrange the unknown } X = \begin{pmatrix} Y^1 \\ Y^2 \\ \vdots \\ Y^{M+1} \\ \Lambda^1 \\ \Lambda^2 \\ \vdots \\ \Lambda^{M+1} \end{pmatrix}.$$

System (4) rewrites as

$$AX = F. \quad (5)$$

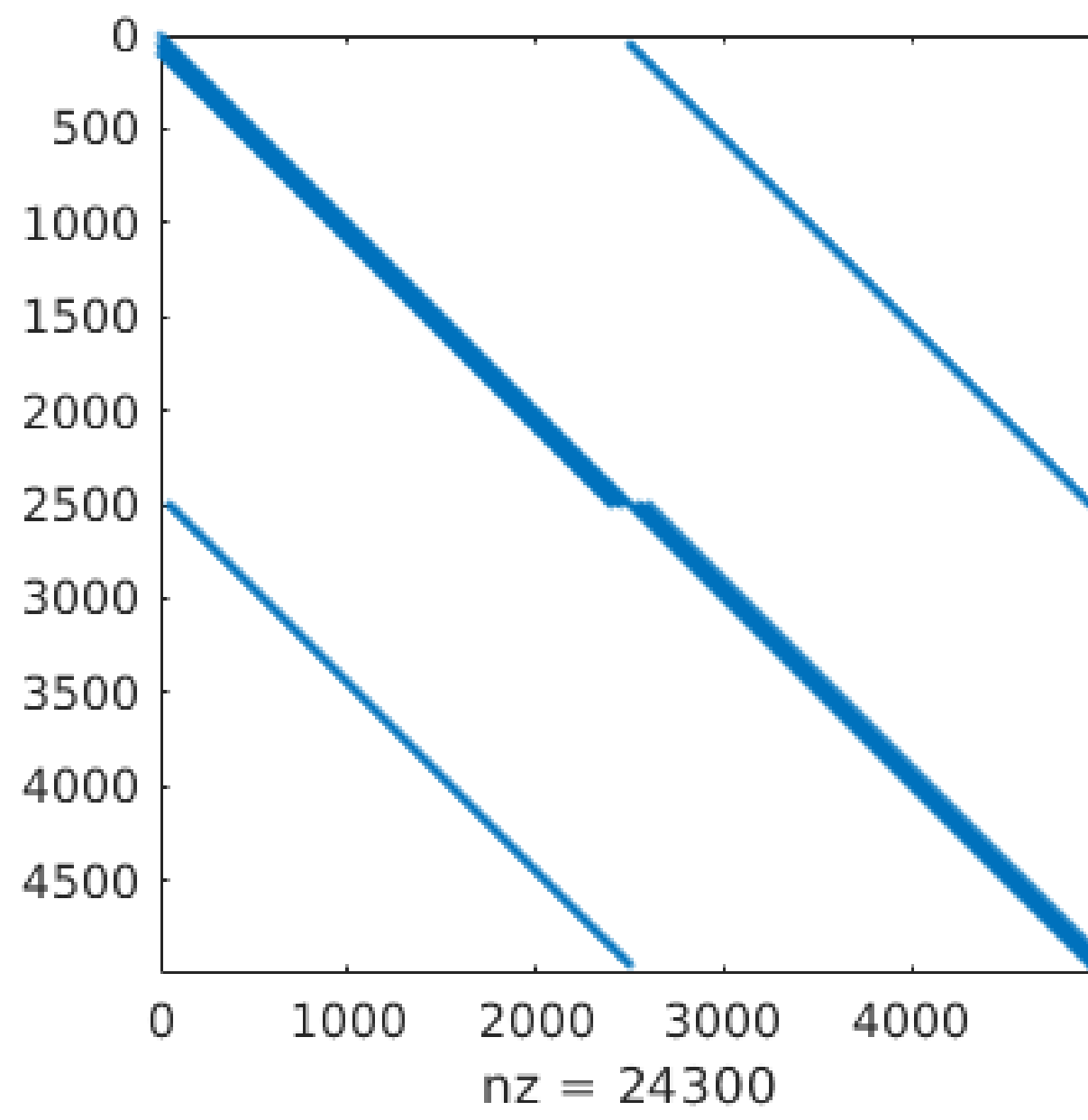


Figure 2: Matrix form of A

Remark. Matrix A is not triangular (coupling between λ and y).

Conjecture 1. The Discrete Problem (4) is well-posed. Assume that $\Delta t = c\Delta x$ where $c \leq 1$. Denote y_h and λ_h is the solution while solving the system (4). There exists a constant $C > 0$ such that ($h = \Delta x$ is a mesh size in space).

$$\begin{cases} \|y_h - y\|_{L^2(\Delta t, \Delta x)} \leq Ch^2, \\ \|y_h - y\|_{L^\infty} \leq Ch^2, \\ \|\lambda_h - \lambda\|_{L^2(\Delta t, \Delta x)} \leq Ch^2, \end{cases} \quad (6)$$

where

$$\|v\|_{L^2(\Delta t, \Delta x)}^2 = (\Delta t)(\Delta x) \sum_{k=1}^{M+1} \sum_{j=1}^{N+1} |v(x_j, t_k)|^2, \quad (7)$$

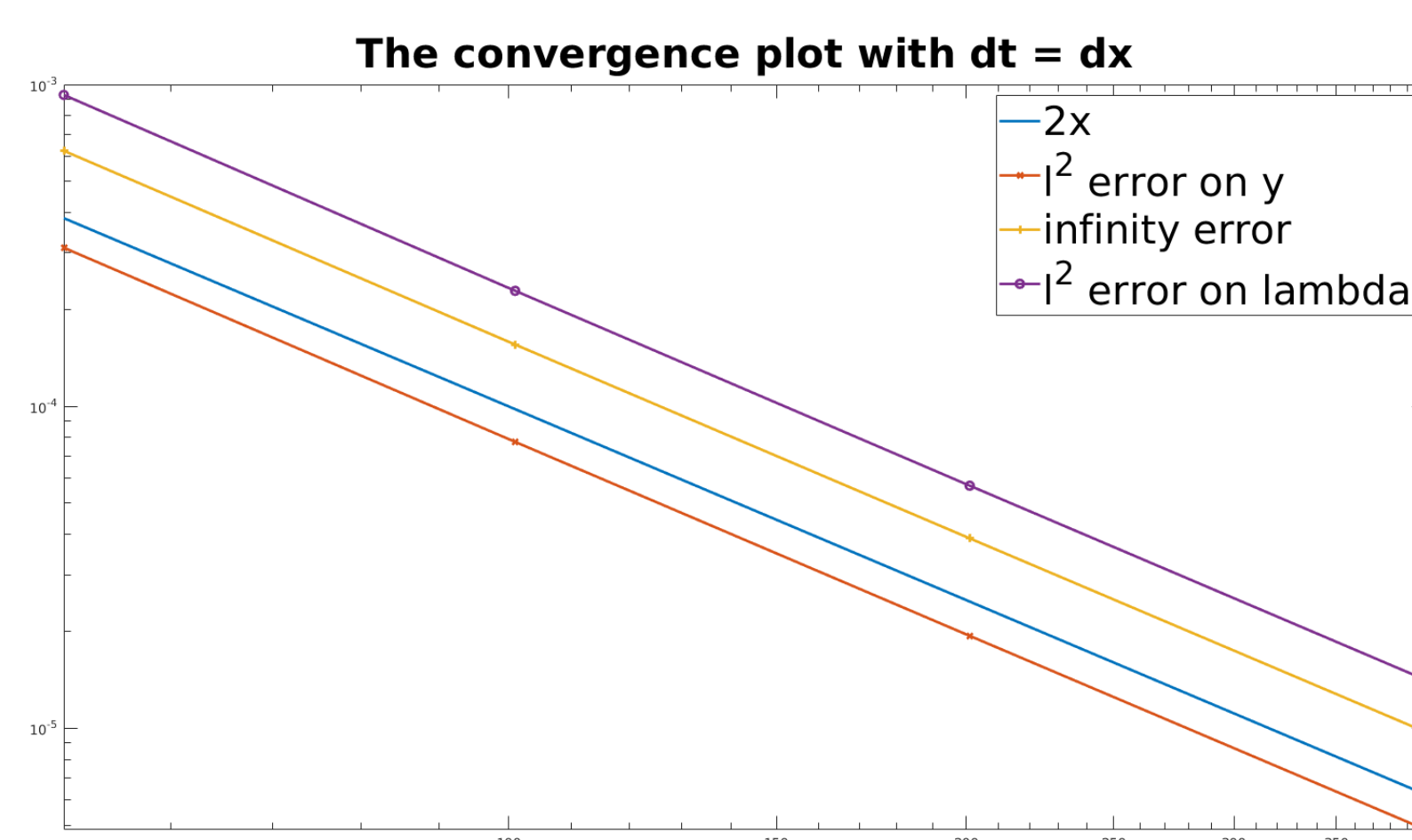
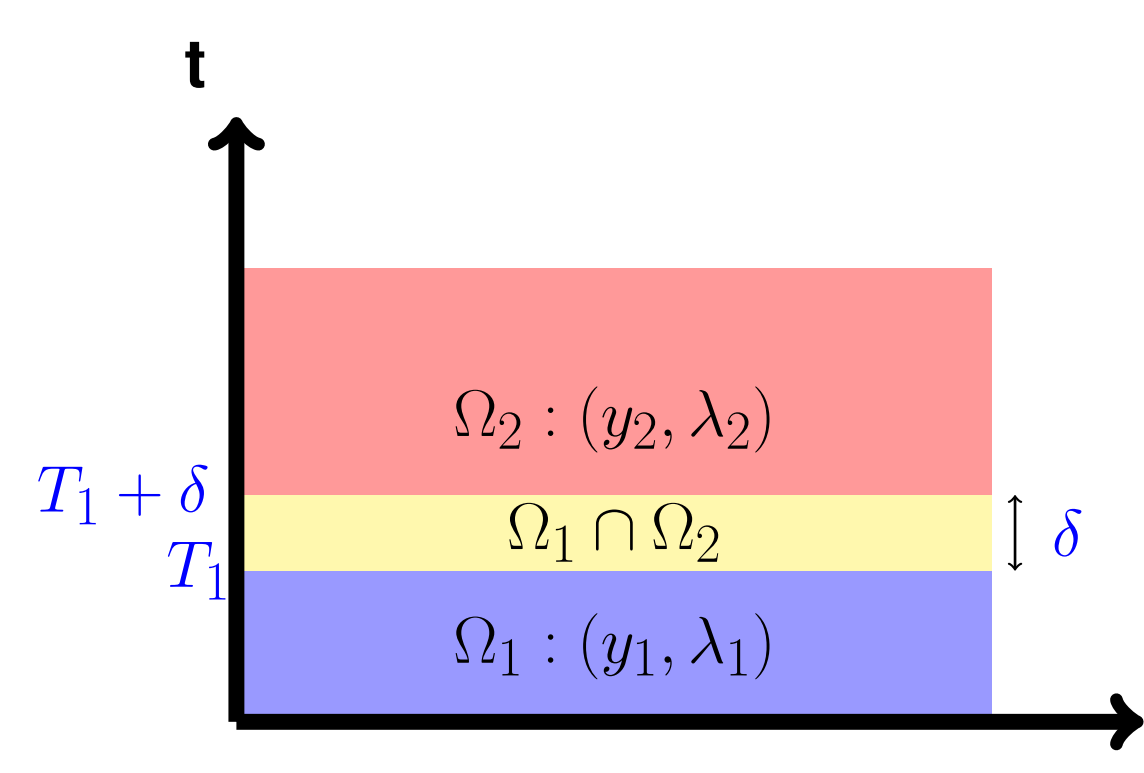


Figure 3: Error with respect to number of mesh point in space $N+1$ when $\Delta t = \Delta x = h = \frac{1}{N}$.

3. Optimized Schwarz time domain decomposition



Schematic representation of the two subdomains $\Omega_1 =$

$\Omega \times (0, T_1 + \delta)$ and $\Omega_2 = \Omega \times (T_1, T)$ ($\gamma' \geq 0$).

$$\begin{cases} \lambda_1^{n+1}(T_1 + \delta) = \lambda_2^n(T_1 + \delta), \\ \lambda_{1,t}^{n+1}(T_1 + \delta) - \gamma' y_1^{n+1}(T_1 + \delta) = \lambda_{2,t}^n(T_1 + \delta) - \gamma' y_2^n(T_1 + \delta) \\ y_2^{n+1}(T_1) = y_1^n(T_1), \\ y_{2,t}^{n+1}(T_1) = y_{1,t}^n(T_1) \end{cases} \quad (8)$$

Theorem 3.1. The time domain decomposition algorithm with transmission condition (8) is well defined.

Convergence factor. Investigate the error with respect to frequency k .

• Fourier transform in space $(\lambda, y) \xrightarrow{\mathcal{F}} (\hat{\lambda}, \hat{y})$.

• Define $(\hat{\lambda}_0)^n = \hat{\lambda}_1^n(k, 0)$, $(\hat{\lambda}'_0)^n = \hat{\lambda}'_{1,t}(k, 0)$, $(\hat{y}_T)^n = \hat{y}_2^n(k, T)$ and $(\hat{y}'_T)^n = \hat{y}'_{2,t}(k, T)$.

There exists a matrix $M(k)$ such that

$$\begin{pmatrix} (\hat{\lambda}_0)^{n+2} \\ (\hat{\lambda}'_0)^{n+2} \end{pmatrix} = M(k) \begin{pmatrix} (\hat{\lambda}_0)^n \\ (\hat{\lambda}'_0)^n \end{pmatrix} \quad (9)$$

Convergence factor $\rho(k) = \text{spectral radius of } M(k)$ and $\rho = \max_{k \in [k_{min}, k_{max}]} \rho(k)$.

Conjecture 2. Considering the time Domain Decomposition Coupled Wave equation, fix $T, T_1, \alpha = 1$, if $\gamma' = \gamma$ for every possible overlapping size $\delta \geq 0$, there always exists a γ such that the convergence factor $\rho(k, \gamma) > 1$ for some frequencies k .

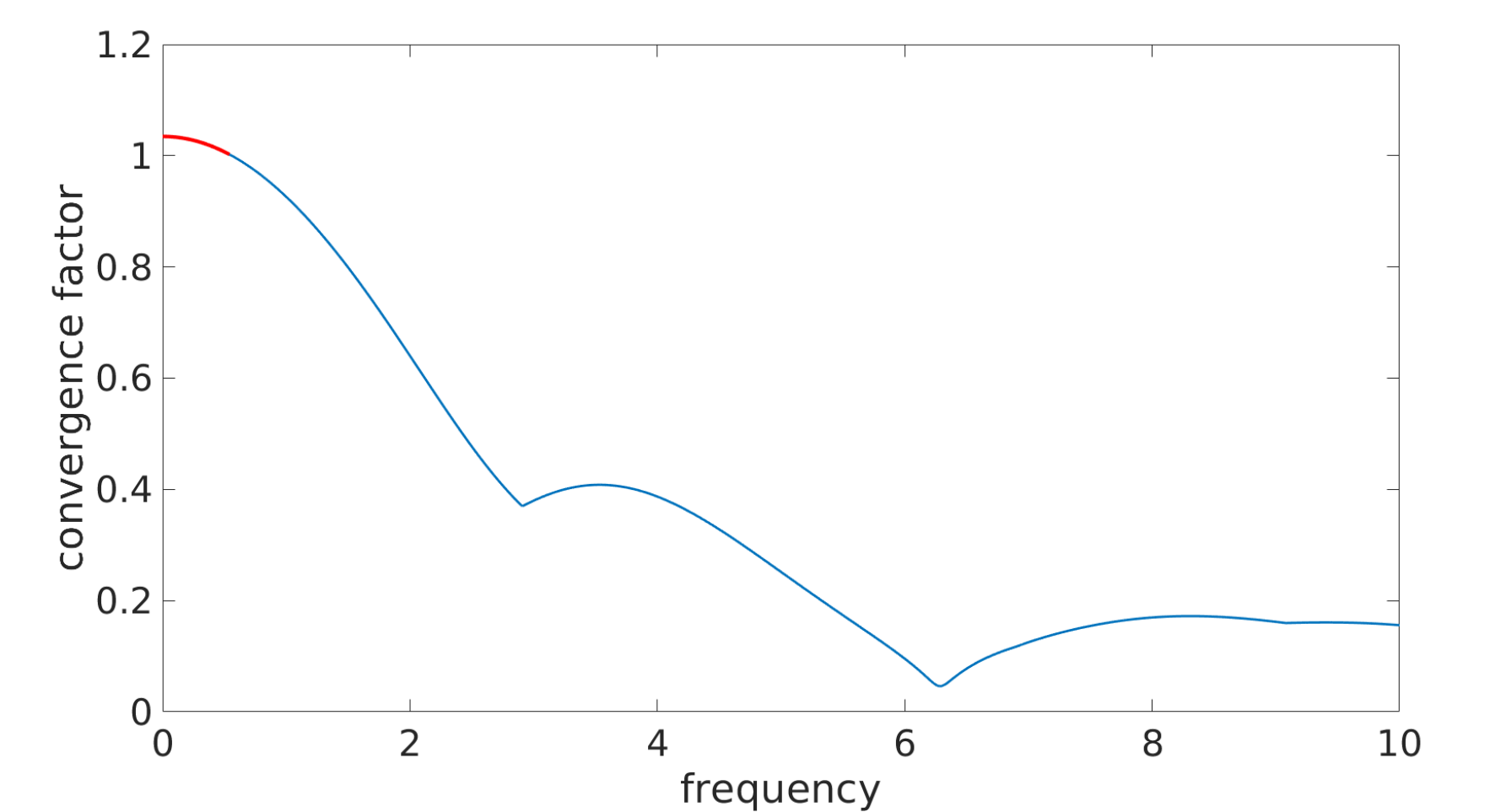


Figure 5: Convergence factor calculation for $T = 1$, $T_1 = 0.45$, $\delta = 0.05$, $\alpha = 1$ and $\gamma = 10$ with frequency k varies from 0 to 10. $\rho(k) > 1$ for k small

Numerical observation. Considering the time domain decomposition system with transmission condition (8), fix $T, T_1, \alpha = 1$ with the overlapping size $\delta \geq 0$. For each $\gamma \geq 0$, there exists a $\gamma'_{max} > 0$ such that for every $\gamma' \leq \gamma'_{max}$, the convergence factor $\rho < 1$, otherwise $\rho > 1$ when $\gamma' > \gamma'_{max}$. Furthermore, there exists one $\gamma'_{opt} > 0$ where convergence factor is minimum over a range of frequency.

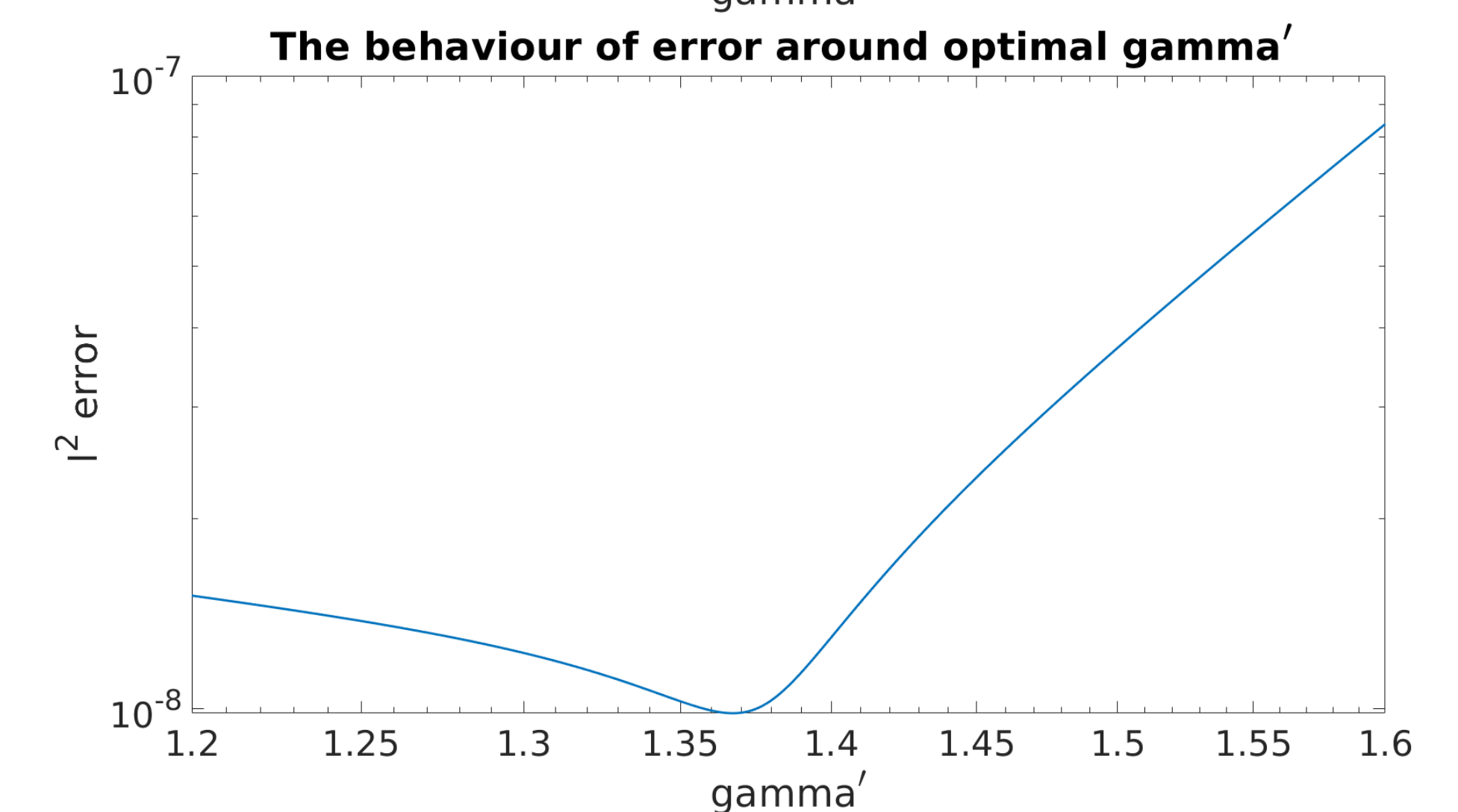
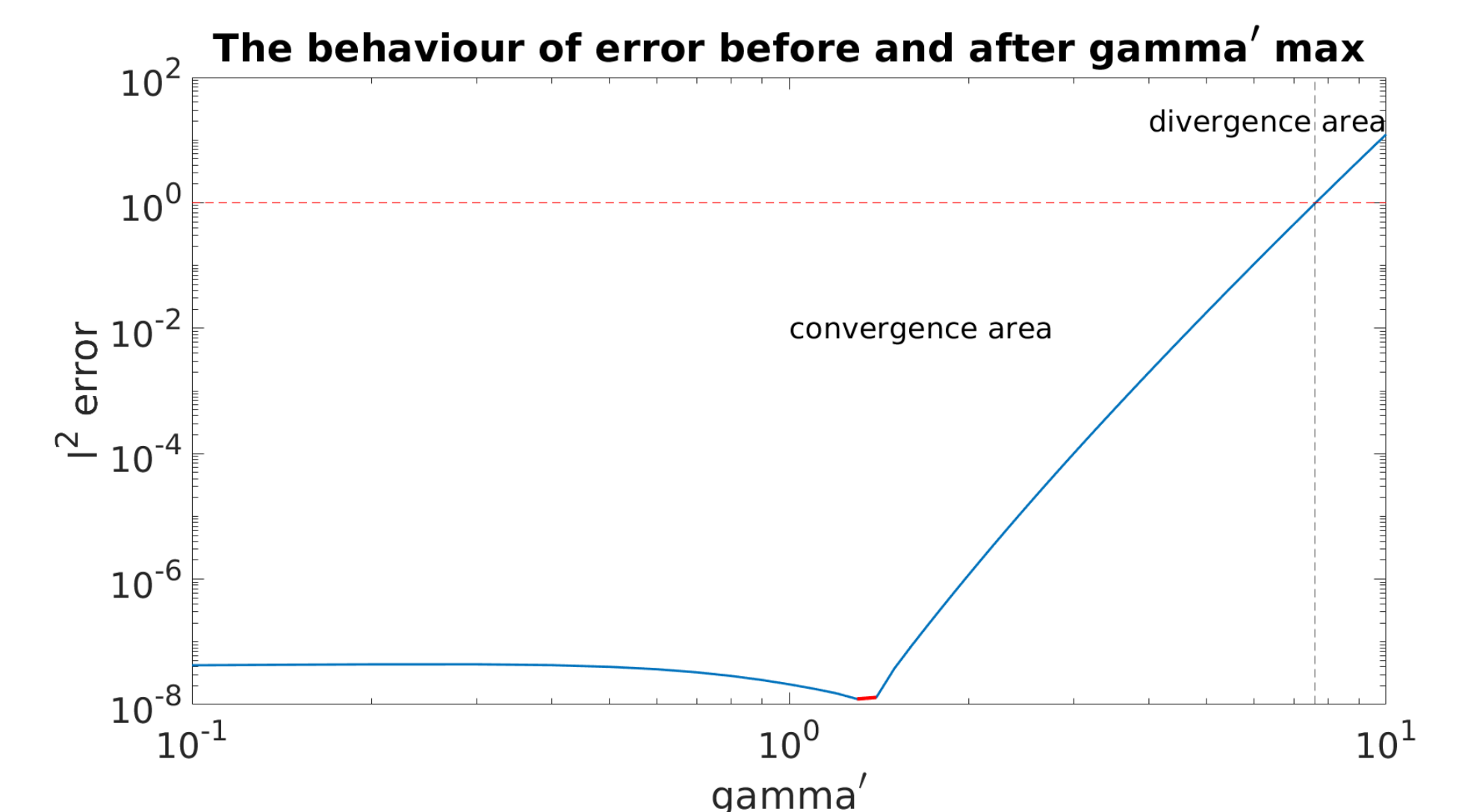


Figure 6: Two figures represent the l^2 error after 20 iterations with respect to γ' where $\gamma = 1$. First figure shows the error when $\gamma' < \gamma'_{max}$ and $\gamma' > \gamma'_{max}$ ($\gamma'_{max} \approx 7$). The second figure is a zoom on optimal parameter $\gamma'_{opt} \approx 1.4$ from $\gamma' = 1.2$ to 1.6

References

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